

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B.Tech.(IT) Theory Examination****September-2018****MA206.01 Statistical and Numerical Methods****Date: 26.09.2018, Wednesday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70**

Instructions: 1) The question paper comprises two sections. 2) Section I and II must be attempted in separate answer sheets. 3) Make suitable assumptions and draw neat figures wherever required. 4) Use of scientific calculator is allowed. 5) Candidate must write exam seat number on paper.

SECTION - I**Q-1 Fill in the blank by choosing the most appropriate answer from the given options.**

- (a) Let A and B are two non-empty subsets of Sample space S , then $P(A \cup B) = P(A) + P(B)$ if **01**
- (i) $A \cup B = S$ (ii) $A \cap B = \phi$
 (iii) $A \subset B$ (iv) $A \cap B \neq \phi$
- (b) To compute rank correlation between two variables X and Y , values of X and Y must be **01**
- (i) ranked (ii) arranged in ascending order
 (iii) positive (iv) numerical quantities
- (c) The average of 7 numbers 7,9,12, x ,5,4,11 is 9. The missing number is **01**
- (i) 13 (ii) 15
 (iii) 14 (iv) 8
- (d) Which of the following is not a measure of central tendency? **01**
- (i) Arithmetic mean (ii) Mode
 (iii) Median (iv) Standard deviation
- (e) Let X denotes number of heads observed in three tosses of a fair coin, then X takes on the values **01**
- (i) $\{0,1\}$ (ii) $\{1,2,3\}$
 (iii) $\{0,1,2\}$ (iv) $\{0,1,2,3\}$
- (f) Suppose a discrete random variable X has the probability distribution **01**
- $P_X(1) = \frac{1}{2}, P_X(2) = \frac{1}{4}, P_X(3) = \frac{1}{8}$ and $P_X(4) = \frac{1}{8}$ then $P(X \leq 1) =$
- (i) $\frac{3}{8}$ (ii) $\frac{7}{8}$
 (iii) $\frac{3}{4}$ (iv) $\frac{1}{2}$
- (g) A random variable $X \sim N(\mu, \sigma^2)$ then **01**
- (i) $P(X > \mu) = 0$ (ii) $P(X < \mu) = 1$
 (iii) $P(X \leq \mu) = 0.5$ (iv) $P(-\infty < X < +\infty) = 0$

Q-2 Attempt any four.

- (a) Students in a statistics class were asked to report the number of pets in their family. **04**
The following data were collected:

3	3	2	4	4	4	3	2	3	2	2	3	4	2	2
2	3	4	4	2	3	2	3	3	2	4	3	3	3	2
2	2	1	4	4	2	3	3	1	2	2	2	2	3	1

Find the frequencies of 1, 2, 3, 4 and compute mean and variance from the frequency distribution obtained by you.

- (b) An urn contains four balls numbered 1,2,3, and 4. One ball is to be drawn at random from the urn. Let the events A, B, and C be defined by $A = \{1,2\}$, $B = \{1,3\}$, and $C = \{1,4\}$. Then show that A,B and C are pair wise independent. **04**
- (c) Bowl B_1 contains two red and four white chips, bowl B_2 contains one red and two white chips, and bowl B_3 contains five red and four white chips. Define the events B_1, B_2, B_3 respectively that bowl $B_i, i = 1,2,3$ is chosen at random. Assume that $P(B_1) = \frac{1}{3}, P(B_2) = \frac{1}{6}$ and $P(B_3) = \frac{1}{2}$. The experiment consists of selecting a bowl with these probabilities and then drawing a chip at random from that bowl. Suppose the red chip is drawn, then compute the probability that it was drawn from bowl B_1 . **04**
- (d) The number of telephone calls arriving at a switchboard during any 10-minute period is known to be a Poisson r.v. X with parameter $\lambda = 2$. **04**
(i) Find the probability that more than three calls will arrive during any 10-minute period.
(ii) Find the probability that no calls will arrive during any 10-minute period.
- (e) Let X denote the score on a preliminary test and Y the score on the final examination of 10 students in mathematics. Assuming linear relationship compute the simple correlation coefficient between the variables X and Y and write your interpretation about relationship between X and Y . You are given $n = 10, \sum X = 683, \sum Y = 813, \sum XY = 56089, \sum X^2 = 47405$ and $\sum Y^2 = 66731$. **04**
- (f) Let $X \sim B(n, p)$ with $n = 5, p = 0.4$. Compute $P(X = 0)$ and $P(X = 1)$. **04**

Q-3 (a) Attempt any two.

- (i) Assume that the length of a phone call in minutes is random variable X with probability density function $f(x) = \begin{cases} \frac{1}{10} e^{(-\frac{1}{10})x}, & x > 0 \\ 0 & \text{otherwise} \end{cases}$. If someone arrives at a phone booth just before you arrive, find the probability that you will have to wait less than 5 minutes. **03**
- (ii) If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$. Find $P(2 < X < 5)$. You are given $P\left(Z < \frac{2}{3}\right) = 0.7475$ and $P\left(Z < \frac{1}{3}\right) = 0.6306$ **03**

- (iii) At a gas station, 180 drivers were asked to record the mileage of their cars and the number of miles per gallon. The results are summarized in the table. **03**

	Sample Mean	Standard deviation
Mileage	24598	14634
Miles per gallon	23.8	3.4

The sample correlation coefficient is $r = -0.17$. Compute the least squares regression line which describes how the number of miles per gallon depends on the mileage

- Q-3 (b)** Assume that A and B are two events with $P(A) > 0, P(B) > 0$ Answer following questions **06**

- (i) Find $P(A|B)$ if (i) $A \cap B = \phi$ (ii) $A \subset B$ (iii) $B \subset A$,
(ii) Show that if $P(A|B) > P(A)$ then $P(B|A) > P(B)$.

OR

- Q-3(b)** Consider the experiment of throwing two fair dice. Find the probability of the event that **06**

- (i) The two faces are same.
(ii) The two faces are same given that the sum is not greater than 3.

SECTION – II

- Q-4** Fill in the blank by choosing the most appropriate answer from the given options.

- (a) The n^{th} difference of a polynomial of degree $n-1$ is _____. **01**
(i) 0 (ii) n
(iii) $n-1$ (iv) $n-2$
- (b) The interpolating polynomial $y = x^3 + 2x + 1$ passes through the point _____. **01**
(i) (0, 1) (ii) (1, -2)
(iii) (3, 1) (iv) (2, 5)
- (c) The relationship between E and ∇ is $\nabla =$ _____. **01**
(i) $1 - E^{-1}$ (ii) $1 - E$
(iii) $1 + E^{-1}$ (iv) $1 + E$
- (d) The average rate of convergence in Bisection method to find real root is _____. **01**
(i) 0.25 (ii) 0.5
(iii) 0.75 (iv) 1

- (e) The first approximation y_1 of the initial value problem $y' = x^2 + y^2$, $y(0) = 0$ obtained by Picard's method is _____. **01**

(i) x

(ii) $\frac{x^2}{2}$

(iii) $\frac{x^3}{3}$

(iv) $\frac{x^4}{4}$

- (f) Basic condition for Bisection method to be applicable in finding root $\alpha \in (a, b)$ of an equation $f(x) = 0$ is _____. **01**

(i) $f(a)f(b) < 0$

(ii) $f(a)f(b) = 0$

(iii) $f(a)f(b) > 0$

(iv) $f(a) = f(b)$

- (g) If interval (a, b) be divided into n equal parts of length h , then $h =$ _____. **01**

(i) $b - a$

(ii) $bn - an$

(iii) $\frac{b-a}{n}$

(iv) $\frac{n(b-a)}{b+a}$

Q-5 Attempt any four.

- (a) Construct Newton's forward interpolation polynomial for the following data set: **04**

x	4	6	8	10
y	1	3	8	16

Use it to find value of y at $x = 4.5$.

- (b) Using Newton's divided difference interpolation formula find interpolating polynomial for the following data set: **04**

x	-1	0	1	3
y	2	1	0	-1

- (c) Using the method of least square fit a curve of the form $y = ax^b$ to the data set: **04**

x	2	3	4	5
y	27.8	62.1	110	161

- (d) Find a real root of $x - 2\sin x = 0$ using Newton-Raphson method correct up to three decimal places with initial value of x as $\frac{\pi}{2}$. **04**

- (e) Use Taylor's series method to find y at $x = 0.03$ from $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$. **04**

- (f) Evaluate the integral $\int_4^{5.2} \log_e x \, dx$ using Simpson's $\frac{3^{th}}{8}$ rule with $h = 0.2$. **04**

Q-6 (a) Attempt any two.

- (i) Consider following tabular values

03

x	50	100	150	200	250
y	618	724	805	906	1032

Determine $y(225)$ using Newton's Backward interpolation formula.

- (ii) A curve is drawn to passes through points given by the data set

03

x	1	1.5	2	2.5	3
y	2	2.4	2.7	2.8	3

Find area bounded below the curve y and above x -axis between $x = 1$ and $x = 3$ using Trapezoidal rule.

- (iii) Find real root of an equation
- $xe^x - 2 = 0$
- using Regula-Falsi method in interval
- $(0.8, 0.9)$

03

correct up to three decimal places.

Q-6(b) Do as directed

- (i) Perform three iterations of the Bisection method to obtain a negative real root of an equation
- $x^3 - 7x + 3 = 0$
- in interval
- $(-3, -2)$
- .

03

- (ii) Use Euler method to find
- $y(1.4)$
- given that
- $\frac{dy}{dx} = x\sqrt{y}$
- ,
- $y(1) = 1$
- with
- $h = 0.1$
- .

03**OR****Q-6(b) Do as directed**

- (i) Using the method of least squares, Find an equation of the form
- $y = ax + b$
- that fits the data set:

03

x	0	1	2	3	4
y	1	5	10	22	38

- (ii) Find second approximation of
- y
- at
- $x = 0.1$
- using Picard's method from

03

$$\frac{dy}{dx} = 6 + xy, y(0) = 1.$$
